HUI-JU FAN

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huijufan.github.io

EDUCATION

The University of Texas at Dallas May 2026
Master of Science, Marketing (STEM-designated) | GPA 3.3

Soochow University, Taiwan June 2024
Bachelor of Business Administration, International Business | GPA 3.9
National Science Council Undergraduate Research Scholarship

RESEARCH EXPERIENCE

The University of Texas at Dallas August 2026 – Present
Research Assistant | Advisor: Professor Yi Ding

- Building a portable electronic nose that pairs a BME688 MOX gas sensor with an ESP32-S3 microcontroller for real-time volatile compound sensing
- Paired the device with an ARCore-tracked smartphone to locate odor sources without exhaustively mapping the space, benchmarked against established gas distribution mapping methods

MIT EECS & CSAIL March 2025 – February 2026
Research Collaborator, Perceptual Engineering | Advisor: Professor Jas Brooks | Remote

- Developed a chemical safety system using electronic tongue technology to detect hazardous interactions between household cleaning products (e.g., bleach and toilet bowl cleaner producing chlorine gas)
- Built a dataset of 44 cleaning products characterized by multiple ion-selective electrode measurements for training a chemical identification model
- Designed and implemented sensor array circuitry using Arduino-based microcontrollers interfaced with multiple ion-selective electrodes for real-time chemical detection
- Applied dimensionality reduction (PCA, t-SNE) in Python to surface discriminative markers despite high ion-signature similarity across surfactant-based products

National Science Council, Taiwan July 2023 – March 2024
Undergraduate Researcher | Consumer Psychology and Linguistic Persuasion

- Analyzed the effect of homophonic puns in idiomatic phrases on perceived playfulness and product attractiveness, yielding a 40% increase in playfulness perception
- Investigated the impact of homophonic puns on information processing fluency, demonstrating a 31% improvement in processing efficiency
- Designed and ran experimental studies with 200+ participants; analyzed results in SPSS and R
- Presented findings to the faculty committee and received the National Science Council Scholarship

PUBLICATIONS & SUBMISSIONS

Fan, H.-J., Yan, X., & Ding, Y. Demo: Osphra: Finding Odor Sources without Mapping Every Point. ACM MobiCom 2026. (Accepted)

PATENTS

Hui-Ju Fan et al. “Osphra: Finding Odor Sources without Mapping Every Point.” U.S. Patent Application, Patent Pending, 2026.

ENTREPRENEURIAL EXPERIENCE

PerVenture: VR-Olfactory Shopping Platform October 2022 – October 2024
Co-Founder & Team Leader | 5–6 member cross-functional team

- Founded and led a startup developing a VR-based olfactory interface for immersive fragrance shopping, addressing the technical gap in remote scent transmission
- Designed and prototyped 3D virtual environments in Unity and C# for one-on-one personalized perfume consultation
- Identified the lack of olfactory display hardware as the critical barrier, which motivated the shift toward graduate research in perceptual engineering and human-computer interaction

SKILLS & ADDITIONAL INFORMATION

Programming: Python (data analysis, signal processing, machine learning), R (statistical analysis), C# (Unity/VR), MATLAB, C/C++ (Arduino & ESP32 firmware)

Hardware & Prototyping: ESP32-S3, Arduino, BME688 MOX gas sensors, ion-selective electrodes, sensor array integration, circuit design, 3D modeling and printing

Data Analysis & Visualization: PCA, t-SNE, multivariate analysis, SPSS, Tableau

Research Methods: User research, experimental design

Languages: Mandarin (native), English (professional), Japanese (conversational)

Work Eligibility: Eligible to work in the U.S. for internships and full-time employment for up to 36 months